KOMTAS

INDUSTRY DATA MODELS

Industry-specific enterprise data model and EDW for banking, investment banking, telco and insurance: prepared subject areas, KPI sets and loading standards

Start with a Model Fit Assessment, implement the model subject area by subject area, then extend it as new needs arise

1 **The challenge**
Symptoms, and what an industry model gives. — p. 2 – 3

2 **Four industries**
Overview, banking, investment banking, telco, insurance. — p. 4 – 8

3 **How it works**
Tiers, fit assessment, fit/gap example, implementation. — p. 9 – 12

4 **Experience and next step**
References, customer side, about Komtaş, next step. — p. 13 – 14

Symptoms: When a Data Warehouse Needs an Industry Model

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Data warehouses often grow one datamart at a time. Without a shared model, definitions drift, new reports take long and regulatory requests turn into projects.

01 · SYMPTOMS

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Every datamart its own definitions

Customer, product and revenue are defined differently in each datamart.

Same KPI, different numbers

Reports from different teams show different values for the same KPI.

Slow new reports

Each new report needs new loads and new joins across many sources.

Regulatory pressure

Regulatory requests need data from many systems, collected by hand each time.

Point-to-point loads

Sources are loaded straight into datamarts; the same data is loaded several times.

Missing history

Changes are overwritten; past states cannot be reported.

New core system or merger

A new core system or a merger needs a common target for data from both sides.

Platform migration

A move to a new or cloud platform is a chance to rebuild on a model instead of copying.

WHAT AN INDUSTRY MODEL CHANGES

One model
shared entities and definitions

KPIs defined once
used by every datamart

History kept
consistent historisation rules

Faster new areas
prepared subject areas

Ready for regulation
data in one place, traceable

What an Industry Data Model Gives

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A prepared, industry-specific enterprise data model «MODEL»: shared core entities, subject areas, KPI sets and loading rules that are adapted rather than designed from scratch.

02 · WHAT IT GIVES

WHAT IS INCLUDED

«DIGITALLENCY DOĞRULASIN»

Shared core
Party, product, time and organisation entities used by every subject area.

Subject areas
Entity and relationship sets per business area, ready to adapt.

KPI sets
KPI definitions per industry, linked to model entities and attributes.

Historisation
History rules, full and delta loads defined once for all tables.

Definitions and standards
Table and column definitions; naming and loading standards.

INDUSTRIES

Banking | Investment banking | Telco | Insurance

WHERE THE MODEL SITS

Sources
Core systems, channels, external and partner data.

EDW on the model
Industry model implemented on the target platform «PLATFORM».

Datamarts and KPI sets
Subject-oriented datamarts with KPIs defined once.

Reporting and analytics
BI, regulatory reporting, analytics and AI use cases.

OUT OF SCOPE

«DIGITALLENCY DOĞRULASIN»

- Changes to source systems
- Full rebuild of BI reports
- Platform licences
- Model licence fees «FIELD»

Where this work stops. Regulatory reporting work builds the reporting chain itself (common regulatory layer, data demand layer, data quality and validation for reports); this package builds the industry model and EDW that can feed it. A DWH assessment evaluates the whole platform and the target state; this package implements an industry model once the target is set.

Four Industries, One Method

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Each industry has its own subject areas, KPI sets and regulatory context. The way of working is the same for all four.

03 · FOUR INDUSTRIES

Banking

Banks and participation banks

SUBJECT AREAS

Party, event, channel, campaign, media, contract; further areas such as product, account, collateral and general ledger.

KPI SETS

Customer 360, sales and profitability, campaign and marketing.

BDDK

«DIGITALLENCY DOĞRULASIN»

Investment banking

Brokerage and investment firms

SUBJECT AREAS

Customer and account, order and transaction, portfolio and position, instrument, market data, fee and commission, channel.

KPI SETS

Trading volume, commission income, assets under management, active customers, channel share.

SPK · MKK

«DIGITALLENCY DOĞRULASIN»

Telco

Telecom operators

SUBJECT AREAS

Party, event, finance, network, contract, product; further areas such as usage and CDR, and billing.

KPI SETS

Churn, ARPU, usage, campaign response.

BTK

«DIGITALLENCY DOĞRULASIN»

Insurance

Health, P&C, life and pension

SUBJECT AREAS

Domains by line of business, shared DWH domains, and datamarts for customer value, fraud, sales and agencies.

KPI SETS

Renewal, loss and combined ratios, hit ratio, STP, provision speed, earned premium, budget tracking.

SEDDK

«DAMALINK DOĞRULASIN»

One method across industries. Every engagement starts with a fit assessment against the customer's sources, implements the model subject area by subject area, and extends it when new needs arise.

Banking: Subject Areas and KPI Sets

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Industry-specific enterprise data model «MODEL» for banks, including participation banks. Subject areas and KPI sets are adapted to the bank's sources in the fit assessment.

04 · BANKING

SUBJECT AREAS

«DIGITALLENCY DOĞRULASIN»

Party
Customers, prospects, related persons and organisations, with roles.

Event
Transactions and customer events across products and channels.

Channel
Branches, ATMs, internet and mobile banking, call centre.

Campaign
Campaigns, offers, target lists and responses.

Media
Communication media and messages used to reach customers.

Contract
Accounts, loans, deposits and other agreements with the bank.

FURTHER SUBJECT AREAS · CONFIRMED IN THE FIT ASSESSMENT

Product · Account · Collateral · General ledger

KPI SETS

«DIGITALLENCY DOĞRULASIN»

Customer 360
Customer count, product holding, channel use and value per customer.

Sales and profitability
Sales volumes and profitability by product, branch and segment.

Campaign and marketing
Campaign reach, response and conversion.

Risk and regulatory feed
Data prepared for risk and regulatory reporting; scope set in the fit assessment.

REGULATION

«DIGITALLENCY DOĞRULASIN»

BDDK
Model definitions support regulatory reporting to BDDK. For participation banks, interest-free product structures are reflected in the contract and product subject areas.

LINKS TO OTHER WORK

Regulatory reporting and Customer 360
The EDW built on the model can feed a regulatory reporting layer; a single customer view from master data management feeds the Customer 360 KPI set.

Shared core first. Party, product, time and organisation entities are loaded once and reused by every banking subject area, so KPIs across datamarts use the same definitions.

Investment Banking: Subject Areas and KPIs

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Industry-specific enterprise data model «MODEL» for brokerage and investment firms. The subject areas below are a proposed starting set.

05 · INVESTMENT BANKING

SUBJECT AREAS

«DIGITALLENCY DOĞRULASIN»

Customer and account
Investors, accounts and authorised persons.

Order and transaction
Orders, executions and settlements across markets.

Portfolio and position
Holdings and positions by account and instrument.

Instrument
Equities, fixed income, funds, derivatives.

Market data
Prices, rates and indices used for valuation.

Fee and commission
Commissions, fees and their calculation basis.

Channel
Branches, online and mobile trading, agencies.

The subject areas above are a proposed starting set; content and boundaries are confirmed in the Model Fit Assessment.

KPIs

«DIGITALLENCY DOĞRULASIN»

Trading volume
by market, instrument, channel and customer segment

Commission income
by product, channel and customer

Assets under management
portfolio size and change over time

Active customers
customers with transactions or holdings in the period

Channel share
share of orders and volume per channel

REGULATION

«DIGITALLENCY DOĞRULASIN»

SPK and MKK
Model definitions take into account reporting needs towards SPK and MKK; the exact scope is set in the fit assessment.

LINKS TO OTHER WORK

Customer data and governance
Customer and account definitions can be shared with master data and data governance work, so that customer counts match across reports.

One view of customer, order and position. Orders, positions and commissions are linked to the same customer and instrument entities, so trading and income KPIs reconcile across channels.

Telco: Subject Areas and KPIs

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Industry-specific enterprise data model «MODEL» for telecom operators, built on an industry logical data model.

06 · TELCO

SUBJECT AREAS

«DIGITALLENCY DOĞRULASIN»

Party
Subscribers, customers, dealers and partners, with roles.

Event
Calls, sessions, service events and customer interactions.

Finance
Revenue, payments, balances and adjustments.

Network
Network elements, cells, sites and capacity.

Contract
Subscriptions, commitments and contract periods.

Product
Products, tariffs, bundles, devices and add-ons.

FURTHER SUBJECT AREAS · CONFIRMED IN THE FIT ASSESSMENT

Usage and CDR | Billing

KPIs

«DIGITALLENCY DOĞRULASIN»

Churn
churned subscribers and churn rate by segment and product

ARPU
average revenue per user by segment, tariff and period

Usage
voice, data and messaging usage per subscriber

Campaign response
response and conversion by campaign and channel

REGULATION

«DIGITALLENCY DOĞRULASIN»

BTK
Reporting needs towards BTK are reviewed in the fit assessment and mapped to model entities where they apply.

LINKS TO OTHER WORK

Datamart consolidation
Where several datamarts calculate churn or ARPU differently, the model can serve as the consolidation target with one definition per KPI.

Usage at scale. Usage and CDR volumes drive platform sizing; history depth and delta load rules for these subject areas are agreed in the fit assessment «PLATFORM».

Insurance: Domains, Datamarts and KPIs

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Industry-specific enterprise data model «MODEL» for insurers, covering health, property and casualty (P&C), and life and pension business.

07 · INSURANCE

LINES OF BUSINESS AND SOURCE DOMAINS

«DAMALINK DOĞRULASIN»

Health

Policy and group policy, customer, policyholder and insured, coverage, underwriting, product, provision, network, institution, agency, third-party administrator, external data.

Property and casualty

Customer, policy and group policy, policyholder and insured, coverage, claim, product, property, network, service, external data.

Life and pension

Customer, company, employee, communication, policy, policyholder and insured, business object, agent, loss event, reinsurance, commission account, coverage, claim, product, booking basis, location.

External sources: SBM, KPS, EGM, KKB and marketing data (SMS, social media, calls, chat, campaigns).

DWH DOMAINS · SHARED ACROSS LINES OF BUSINESS

«DAMALINK DOĞRULASIN»

Agency · Agent · Animal · Bank · Booking basis · Building · Business object · Business partner · Claim · Coinsurance · Commission account · Communication · Company · Coverage · Customer · Damage · Employee · External · Health · Institution · Insured · Life · Litigation · Location · Loss event · Network · Policy · Policyholder · Product · Property · Provision · Reinsurance · Service · Underwriting · Vehicle

DATAMARTS

«DAMALINK DOĞRULASIN»

- **Customer value management** — one customer across lines of business
- **VIP customer** — value rules on premium, collection, coverage
- **Retail segmentation** — value from earned premium and claims
- **Call centre performance** — IVR, complaints, messaging, NPS
- **Fraud** — internal and customer fraud rules
- **Health** — provision and underwriting performance
- **Sales** — premium by region and agent; plan vs actual
- **Cross-sell and penetration** — products per customer across lines
- **Agency incentive** — agency representative incentive tracking

KPI SET

«DAMALINK DOĞRULASIN»

New, lost, existing customers · Policies by status · Renewal rate · Loss ratio · Combined ratio · Hit ratio (quote to policy) · STP rate · Provision speed · Exposure · Earned premium · Budget tracking: NEP, GLR, NLR, CR, CoR · MTD and YTD views

Regulation and customer data. SEDDK reporting needs are reviewed in the fit assessment. A single customer view across lines of business feeds the customer value and cross-sell datamarts; analytics such as complaint analysis, geographic risk and renewal prediction use the same domains. «DAMALINK DOĞRULASIN»

Approach: Fit, Implement, Extend

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The Model Fit Assessment measures how well the model fits the customer's sources. Implementation follows subject area by subject area. Extension adds new areas later.

08 · APPROACH AND TIERS

A Model Fit Assessment

fixed fee · 4 – 8 weeks by size

- Landscape: sources, DWH, datamarts, KPIs
- Fit mapping to subject areas and KPI sets
- Gap analysis and fit ratio per subject area
- Adapted logical model draft and roadmap

OUTPUT AND ACCEPTANCE

Fit/gap accepted · fit/gap matrix · fit ratio table · logical model draft · KPI mapping · B roadmap and estimate

«DIGITALLENCY DOĞRULASIN»

B Model Implementation

foundation once, then per subject area

- Physical model for the target platform
- Core entities, standards, ETL framework
- Per subject area: mapping, ETL, KPI set
- Parallel validation against current reports

OUTPUT AND ACCEPTANCE

Per subject area: reconciliation with agreed KPIs within tolerance «NUMBER» · loads running · handover

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C Extension

per item · optional monthly service

- C1 New subject area, KPI set or source
- Same method and acceptance as B
- C2 Monthly run and model maintenance
- Or run under an operations support service

OUTPUT AND ACCEPTANCE

Per item: same acceptance as B · C2 monthly report «OPTIONAL»

«DIGITALLENCY DOĞRULASIN»

INDICATIVE TIMELINE

A Fit Assessment 4 – 8 weeks → **B Implementation** foundation, then subject areas in waves → **C Extension** per item, or monthly

Fit assessment first. Implementation starts from the Model Fit Assessment, or from an equivalent assessment already done by the customer «A REF». Each tier is proposed separately.

Model Fit Assessment: Flow and Timeline

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Five phases from kick-off to an adapted logical model and an implementation plan. The industry and the subject areas in scope are agreed at kick-off.

09 · MODEL FIT ASSESSMENT

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P0 Kick-off

industry and scope

- Industry branch
- Subject areas in scope
- Sources and contacts
- Plan and access

P1 Landscape

sources, DWH, reports

- Source systems and volumes
- Existing DWH and datamarts
- Reports and KPI definitions
- Business owner interviews

P2 Fit mapping

sources to the model

- Source to model entities
- Attributes to model attributes
- KPI needs to model KPI sets
- Evidence per subject area

P3 Gap analysis

gaps and fit ratio

- In the model, not in sources
- In sources, not in the model
- Customisation needs
- Fit ratio per subject area

P4 Target and plan

model draft and roadmap

- Adapted logical model draft
- Prioritised subject areas
- Platform choice «PLATFORM»
- B roadmap and estimate

DELIVERABLES

«DIGITALLENCY DOĞRULASIN»

Fit/gap matrix

Source entities and attributes against the model.

Fit ratio table

Fit ratio per subject area, with main gaps.

Logical model draft

Model adapted to the customer's scope.

KPI mapping

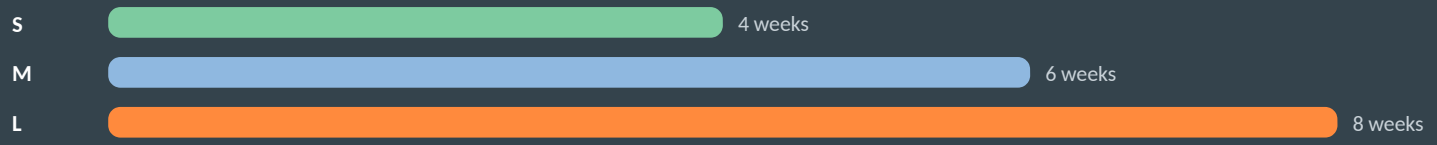
Requested KPIs against model KPI sets.

B roadmap and estimate

Waves of subject areas, effort and plan.

INDICATIVE DURATION BY SIZE

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How size is set

Subject areas in scope, number of source systems, data volume, history depth, number of KPIs and expected customisation.

Fit/Gap: An Illustrative Example

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Example output of a Model Fit Assessment for a bank. For each subject area, source entities are mapped to the model and a fit ratio is calculated.

10 · FIT/GAP EXAMPLE

ILLUSTRATIVE EXAMPLE

Invented values for illustration, not results from a customer.

FIT RATIO PER SUBJECT AREA · BANKING

«DIGITALLENCY DOĞRULASIN»

Subject area	Entities	Fit ratio	Main gap
Party	24	82%	Corporate group hierarchy
Contract	31	71%	Product-specific attributes
Event	18	76%	Digital channel event detail
Channel	9	64%	New digital channel types
Campaign	12	48%	Responses not kept in sources
Media	6	35%	No current source
Overall	**100**	**69%**	*weighted by entities*

70% and above: adopt

50 – 69%: adapt

below 50%: decide

GAP TYPES · NUMBER OF ATTRIBUTES

46 — **In the model, not in sources**
Kept for later sources or left empty by rule.

23 — **In sources, not in the model**
Added as custom attributes in custom tables.

11 — **Customisation needed**
Model entity changed to fit the business.

KPI MAPPING · 120 KPIs REQUESTED



Covered by the model KPI sets

Derived from model entities

New definition needed

Reading: party, event and contract are adopted in the first wave; campaign and media are decided after a source review.

How the fit ratio is read. Fit ratio is the share of in-scope model attributes that can be filled from current sources. Low ratios point to source gaps or customisation; the ratios also set the order of subject areas in implementation.

Implementation by Subject Area

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The foundation is built once. Subject areas then follow in waves, each with its own mapping, loads, KPI set and parallel validation.

11 · IMPLEMENTATION

FOUNDATION · BUILT ONCE

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Physical model

Logical model turned into a physical model for «PLATFORM».

Core entities

Party, product, time and organisation loaded first.

Standards

Naming and loading standards, header attributes, default values.

Historisation

History rules; full and delta loads for all tables.

ETL framework

Parametric loads, delta detection and a recovery strategy.

Quality and semantic base

Data quality checks and a semantic layer base for BI.

PER SUBJECT AREA · IN WAVES

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Mapping — Source-to-target matrix per entity and attribute.

Loads — ETL on the framework, with delta and history.

KPI set — KPIs and a reports base for the subject area.

Parallel validation — Results reconciled with current reports.

Handover — Documents, operations guide, acceptance.

Waves. Subject areas are grouped into waves in the order set by the fit assessment; each wave is accepted on its own. New subject areas after go-live are added as extensions.

Acceptance per subject area. Results reconcile with the agreed KPIs within a tolerance of «NUMBER», and loads run on schedule. «DIGITALLENCY DOĞRULASIN»

Experience and What the Customer Provides

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Industry data model and data warehouse work shown by sector, and the roles and conditions needed on the customer side.

REFERENCE WORK

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Client	Work	Industry	Year
Three investment firms	Investment banking data warehouse	Investment banking	«YEAR»
A telecom group	Enterprise data warehouse on an industry logical data model	Telco	«YEAR»
Two participation banks and a Gulf bank	Banking EDW	Banking	«YEAR»
An insurance group	Customer data across group systems (single customer view) «DAMALINK DOĞRULASIN»	Insurance	«YEAR»
«REFERENCE PROJECT»	Insurance data model implementation	Insurance	«YEAR»

WHAT THE CUSTOMER PROVIDES

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Business owners
one per subject area

- Scope and priorities
- KPI definitions and approval
- Acceptance of results

DWH team
current DWH and datamarts

- Existing models and loads
- Current reports for validation
- Platform and environments

Source system owners
core and channel systems

- Access to sources and metadata
- Explanation of source data
- Change and load windows

CONDITIONS TO START

Access to sources
Read access to source metadata and sample data.

Named business owners
People who define and accept KPIs per subject area.

Current KPI definitions
Reports and definitions used for parallel validation.

Platform direction
Target platform known, or decided in the fit assessment.

References are shown by sector, without customer names.

About Komtaş, and the Next Step

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Who delivers Industry Data Models, and what is needed to start.

12 · ABOUT KOMTAŞ AND NEXT STEP

WHY KOMTAŞ

Data warehouse experience
Data warehouse, data integration and BI projects on platforms such as Teradata and Informatica.

Industry data models in four sectors
Banking, investment banking, telco and insurance; experience on p. 13. «DIGITALLENCY DOĞRULASIN»

Google Cloud and Microsoft Azure partner
Komtaş is a Google Cloud Premier partner and a Microsoft Azure partner.

Customer data and governance
Single customer view and data governance work that use the same model definitions.

From fit assessment to operation
Assessment, implementation, extension and operations support from one partner.

Since 1989, focused on data · 250+ specialists · offices in Istanbul, Ankara, Izmir and Dubai

NEXT STEPS

1. **Walkthrough** — today — you are here
2. **Discovery call** — industry, sources, current DWH
3. **Fit Assessment proposal** — fixed fee, scope and plan
4. **Kick-off** — access, landscape inventory

Platform change as a starting point. A DWH renewal or a move to a cloud data warehouse «PLATFORM» is a natural point to adopt an industry model; the fit assessment can run alongside the platform decision.

Useful for the discovery call. Industry and lines of business; number of source systems and data volume; existing DWH and datamarts; number of KPIs and reports; history depth needed; target platform; regulatory reporting needs.